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G-Bridge: Two-stage QA for viscous dissipation and vortex-stretching from factory sensor data

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Software

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Summary

G-Bridge is an open-source Python pipeline for inline quality assurance (QA) of pipe flows that bridges factory grade proxies-pressure drop, temperature delta, vibration, and line speed to turbulence aware diagnostics. The software follows a two-stage design: (i) a regressor maps proxies to $\log_{10} \phi_v$ (viscous dissipation); (ii) a hurdle model first detects $\gamma > 0$ events and then regresses $\log(1+\gamma)$ on positives to estimate vortex-stretching magnitude. The package provides containers, unit tests, pre-trained checkpoints, and an audit script that regenerates tables and figures from saved predictions with input hash checking, enabling deterministic reproduction across environments.

Methods

Stage 1 regresses $\log_{10} \phi_v$ from factory proxies; Stage 2 uses a hurdle design that (i) detects $\gamma > 0$ and (ii) regresses $\log(1+\gamma)$ on the positives.

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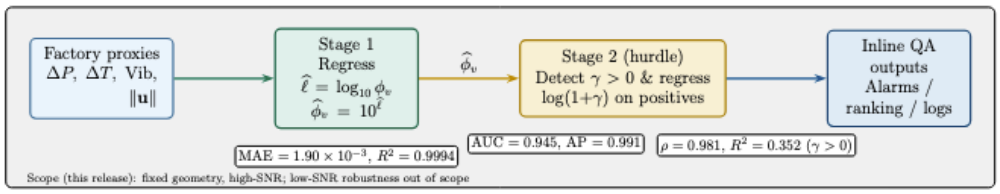

Figure 1: *G-Bridge* maps factory-grade proxies to $\widehat{\phi}_v$ (Stage 1) and screens/ranks γ (Stage 2) for inline QA. Badges show audited test performance on a spatial (file-level) holdout.

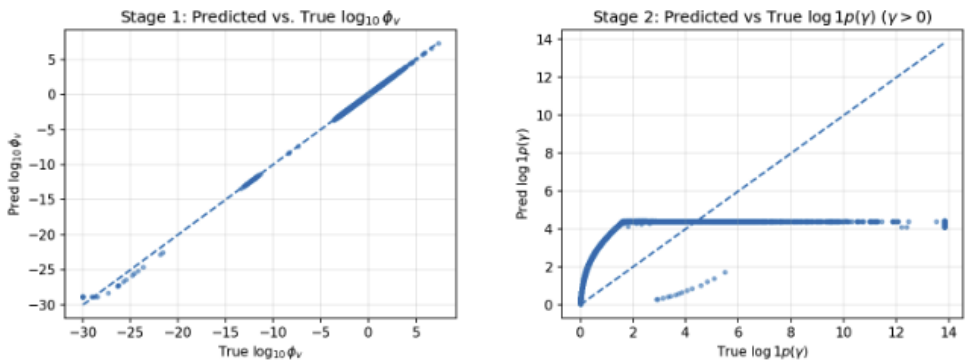

Figure 2: Predicted vs. true for Stage 1 (left: $\log_{10}\phi_v$) and Stage 2 magnitude (right: $\log 1p(\gamma)$, $\gamma > 0$ only). Diagonals indicate perfect prediction; Stage 2 is evaluated on the positive-γ subset per the hurdle protocol.

Figure 1: Overview of G-Bridge and representative results. (A) Two-stage pipeline from factory proxies to physics-aware QA signals. (B) Stage 1 predicted vs. true in $\log_{10}\phi_v$. (C) Stage 2 predicted vs. true on positives in $\log(1+\gamma)$.

Statement of need

QA lines routinely record proxy signals, yet the flow-physics indicators most aligned with product and process health ϕ_v and γ, are not directly instrumented. Offline CFD can recover them but is too slow/costly for inline decisions. **G-Bridge** offers a lightweight, reproducible path from proxies to physics-aligned QA signals: Stage 1 yields a stable dissipation trend operators can threshold/monitor; Stage 2 produces a ranked list of γ - risk locations for targeted inspection. The implementation emphasizes operational fit (CSV/HDF5 I/O, CLI verbs for training/verification/figure regeneration) and auditability (fixed seeds, metrics JSON, and SHA-256 manifests). The current release targets fixed-geometry, high-SNR settings; noise-robust operation and broader geometries are deferred to future work. The code is Apache-2.0 and leverages widely used tooling (e.g., scikit-learn) and OpenFOAM-derived data for examples (Jasak, 2009; Pedregosa et al., 2011).

Acknowledgements

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